



**Politecnico  
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# Battery dimensioning for Renault Zoe

Master's Degree in Automotive Engineering

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# 1 Introduction

Battery dimensioning is a critical aspect of electric vehicle (EV) development, directly influencing performance, range, safety, and overall vehicle integration. This document outlines the key factors in battery dimensioning of a Renault Zoe, a compact and widely popular EV, with a focus on producing a preliminary 3D design of the battery pack. The first design parameters to be set are reported below:

## 1. Electrical Sizing of the battery: Voltage, Nominal Energy

The electrical characteristics of the battery pack are chosen considering typical parameters for B-segment EV cars, as reported in Table 1.

|                 |           |     |
|-----------------|-----------|-----|
| Nominal Voltage | 400 V     | $V$ |
| Energy          | 52.5 kWh  | $E$ |
| Capacity        | 131.25 Ah | $Q$ |

Table 1: Battery pack specifications

## 2. Battery Cell Selection

The chosen battery cell is the LG Chem E63 lithium-ion (Li-ion) pouch cell with a nominal capacity of 65.6 Ah and a nominal voltage of 3.60 V. The specifications of the above-mentioned cell are reported in Table 3.

## 3. Battery Layout

The structural configuration of battery cells selected for the battery pack design is the **Cell-to-Module** arrangement. In this design, the overall characteristics of the battery pack are achieved by interconnecting multiple battery modules. In particular, in this study, the battery pack is composed of 10 modules connected in series. The single battery module specifications are reported in Table 2.

|                  |           |     |
|------------------|-----------|-----|
| Number of module | 10        | $N$ |
| Nominal Voltage  | 40 V      | $V$ |
| Nominal Energy   | 5.25 kWh  | $E$ |
| Capacity         | 131.25 Ah | $Q$ |

Table 2: Cell-to-Module specifications

Other factors, such as gravimetric and volumetric energy density, thermal management system and electrical connections are taken into account in the next sections of this document.

|                            |                                                                                                    |                                                                                                                                                                      |
|----------------------------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Discharge Capacity         | Under 25 °C, 2.50 V - 4.20 V<br>Standard charge / 21.6 A discharge                                 | Nominal 65.6 Ah<br>Minimum 64.6 Ah                                                                                                                                   |
| Standard charge            | Constant current<br>Constant current and voltage<br>End condition (current cut off)<br>Temperature | 21.6 A / 4.05 V<br>13.0 A / 4.20 V<br>3.25 A<br>25 ± 2 °C                                                                                                            |
| Normal charge              | Constant power<br>Termination voltage and power<br>Temperature                                     | 6.25 W<br>4.157 V / 6.25 W<br>10 °C – 50 °C                                                                                                                          |
| 22 kW fast charge          | Constant power<br>Termination voltage and power<br>Temperature                                     | Max. 114.25 W<br>4.157 V / 6.25 W<br>10 °C – 45 °C                                                                                                                   |
| 43 kW fast charge          | Constant power<br>Termination voltage and power<br>Temperature                                     | Max. 224 W<br>4.157 V / 6.25 W<br>25 °C – 45 °C                                                                                                                      |
| Charge at low temperatures | Constant power<br><br>Termination voltage and power<br><br>Temperature                             | Max. 69 W at 0 °C<br>Max. 46.9 W at -10 °C<br>Max. 11.7 W at -20 °C<br>4.157V/13.13 W at 0 °C<br>4.157V/5.26 W at -10 °C<br>4.157V/2.60 W at -20 °C<br>-20 °C – 0 °C |
| Nominal Voltage            | 3.60                                                                                               | V                                                                                                                                                                    |
| Voltage                    | 2.50 – 4.20                                                                                        | V                                                                                                                                                                    |
| Thickness                  | 11.5 ± 0.2                                                                                         | mm                                                                                                                                                                   |
| Weight                     | 964.9 ± 8                                                                                          | g                                                                                                                                                                    |
| Width                      | 125                                                                                                | mm                                                                                                                                                                   |
| Height                     | 325                                                                                                | mm                                                                                                                                                                   |
| Thickness                  | 11.5                                                                                               | mm                                                                                                                                                                   |
| Nominal Capacity at 0 °C   | 57.4                                                                                               | Ah                                                                                                                                                                   |
| Nominal Capacity at 25 °C  | 63.5                                                                                               | Ah                                                                                                                                                                   |
| Nominal Capacity at 45 °C  | 64.8                                                                                               | Ah                                                                                                                                                                   |

Table 3: LG Chem E63's specifications

## 2 Battery cell selection and battery layout

The cell type selected for this battery pack is a pouch cell, which has a cost in between the common cylindrical and the less adopted prismatic one. Its main advantages are high volumetric and gravimetric density due to their size and weight. The flat design allows for even heat distribution, even though it is challenging to well manage the cooling in automotive applications. If the thermal management is not well-designed, the cells starts to swell due to gas generation inside of them, compromising the integrity of the battery back or of the module where they are installed. Moreover, the slim design makes the cell slightly flexible, which increases its mechanical vulnerability compared to other configurations and generally results in a shorter life cycle.

As previously mentioned, the selected cell is a LG Chem E63 lithium-ion (Li-ion) with a nominal capacity of 65.6 Ah and a nominal voltage of 3.60 V, with a battery layout cell-to-module.

This layout enhances maintainability, as replacing a single module is easier and more cost-effective than replacing the entire pack if a pouch cell fails due to swelling, temperature, or mechanical stress. Furthermore, the modular design offers flexibility for battery manufacturers, allowing the same module to be utilized across different vehicle segments. By simply adjusting the number and arrangement of modules, manufacturers can avoid the need to design a new battery pack for each vehicle type. However, the cell-to-module configuration requires additional space to accommodate the connections between modules.

In Figure 1 and Figure 2 the 3D CAD of the cell is shown, along with its dimensions.

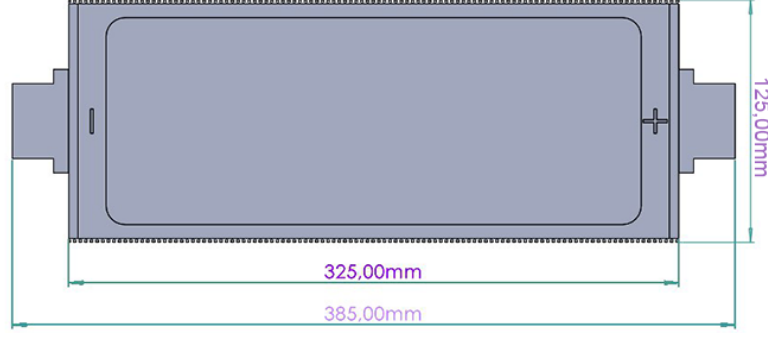


Figure 1: LG Chem *E63* pouch cell front view



Figure 2: LG Chem *E63* pouch cell lateral view

As stated in the introduction, the entire pack is constructed using a **10-module series connection**, where each module consists of an  $11s2p$  configuration of cells. This design provides a nominal capacity of 131.25 Ah, carefully selected to meet the vehicle's performance and range requirements.

The computation of the series-parallel connections  $11s2p$  ( $XsYp$ ) required to match the desired specification of voltage and energy is carried out through the following formulas:

$$X = \frac{\text{Nominal Module Voltage}}{\text{Nominal Cell Voltage}} = \frac{40V}{3.60V} = 11 \quad (1)$$

$$Y = \frac{\text{Nominal Module Capacity}}{\text{Nominal Cell Capacity}} = \frac{131.25Ah}{65.5Ah} = 2 \quad (2)$$

## 2.1 Cells configuration

Once the required number of series and parallel cell connections is determined, the design of the electrical wiring can proceed. For the series connection, the negative terminal of one cell is connected to the positive terminal of the next cell. This type of connection ensures that all cells share the same capacity and current, while the total voltage becomes the sum of the individual cell voltages. On the other hand, parallel connections are used to increase the overall capacity of the battery pack while maintaining the same voltage. This is achieved by connecting cells with matching poles (positive to positive and negative to negative).

To achieve the  $11s2p$  configuration and streamline the cable connections within each module, the cells are arranged as illustrated schematically in Figure 3. In this diagram, the series connections are marked in blue, while the parallel connections are highlighted in red. The series connection is optimized by positioning two consecutive cells in opposite orientations, which aligns the opposite-sign poles (e.g., the positive pole of one cell and the negative pole of the adjacent cell) in close proximity, simplifying the connection process. After connecting pairs of 11 cells in series, the parallel connection is established by linking the negative terminals, which are located at the opposite edges of the module, as well as the positive terminals, which are positioned closer together in the center of the module. Figure 4 illustrates how the  $11s2p$  cells are arranged, using the symbols + for the positive pole and - for the negative pole.

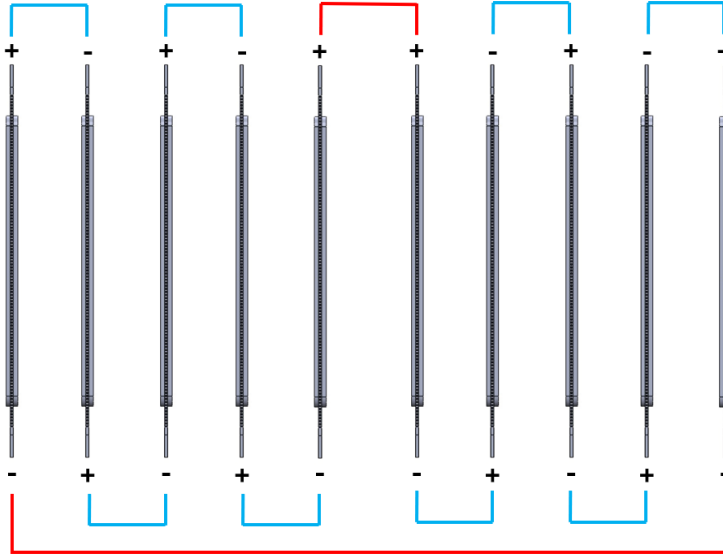


Figure 3: Series-parallel connection sketch

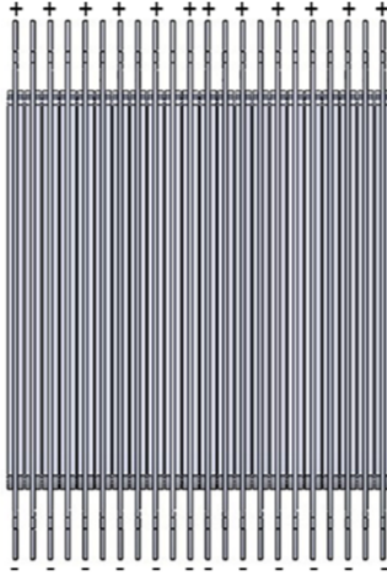


Figure 4: 11s2p cells configuration

It should be noted that no air gap is left between the cells, as pouch cells are typically mounted under pressure. However, their thickness generally expands by approximately 5 – 10% due to the high temperatures and the gases produced within the cells. Despite this expansion, pouch cells are designed to operate under high load conditions. For this reason, they typically exhibit a high volumetric energy density, making them well-suited for demanding applications.

### 3 Thermal management

Considering that batteries operate under demanding working conditions, a cooling system must be taken into account to avoid the swell due to the high temperatures. Various cooling system can be used, such as forced air cooling, surface cooling and edge cooling. The latter is commonly used and is one of the more efficient way to manage the heat transfer for the pouch cells. Considering that pouch cells are made from aluminium and other materials which have a high thermal conductivity, a cold plate is designed to be applied along the longest edge of the cells to dissipate a significant amount of heat. To achieve this, a pump is required to circulate an appropriate fluid through a tube within the cold plate. This type of system is referred to as **passive indirect liquid cooling**, as illustrated in Figure 5, which shows the cold plate developed for the module designed in this survey.

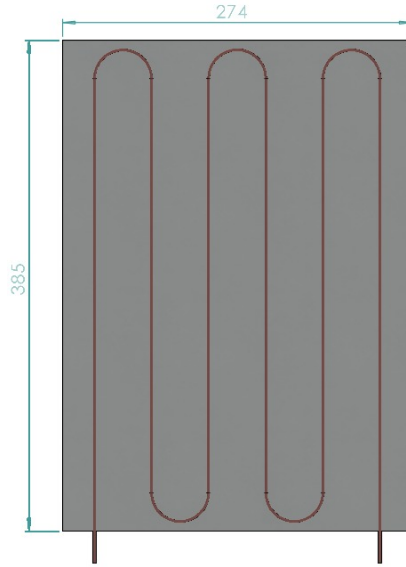


Figure 5: Cold plate - passive indirect fluid cooling system

The coldplate thickness is 5 mm, and the holes through which the tube passes are 3 mm diameter. Width and height are such that the cold plate covers the entire  $11s2p$  cells array, optimizing both the cooling system and the module volume.

## 4 3D-design

To design a 3D model of the discussed module, several factors need to be considered. The cells are connected using busbars, which are bonded to them, and the dimensions of these components must be included in the design. Additional considerations include the integration of the cooling system, a small gap between the cells and the upper case cover, and appropriate tolerances to ensure ease of assembly. Moreover, space must be allocated for the Battery Management System (BMS), which plays a critical role in the battery's operation. The BMS is essential for ensuring the safe and efficient performance of the battery, as well as extending its lifespan. The material chosen for the module is the 6063 aluminium alloy with a density  $d = 2.7 \text{ g/cm}^3$ . Taking all these factors into account, the characteristics of the final module are summarized in Table 4.

|                |                   |               |
|----------------|-------------------|---------------|
| Length         | 385               | mm            |
| Height         | 133               | mm            |
| Width          | 276               | mm            |
| Case Thickness | 2.00              | mm            |
| Volume         | $1.41 \cdot 10^7$ | $\text{mm}^3$ |
| Mass           | 27.2              | kg            |

Table 4: Module case specifications

The total mass includes the weight of the case, cooling system, cells, and all components required for the electrical connections. In Figure 6 and Figure 7, a 3D model of the module is presented along with its dimensions. The positive terminal of the module is positioned at the front, while the negative terminal is located on the opposite side. This arrangement is designed to facilitate the series connection between consecutive modules efficiently.

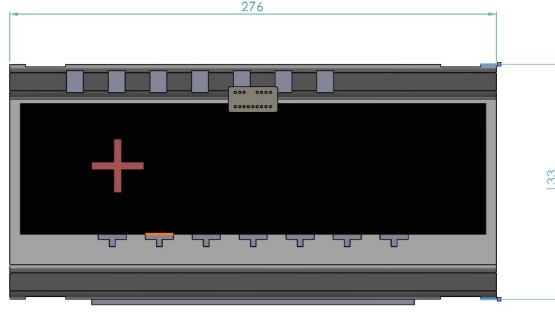


Figure 6: Battery module - Front view

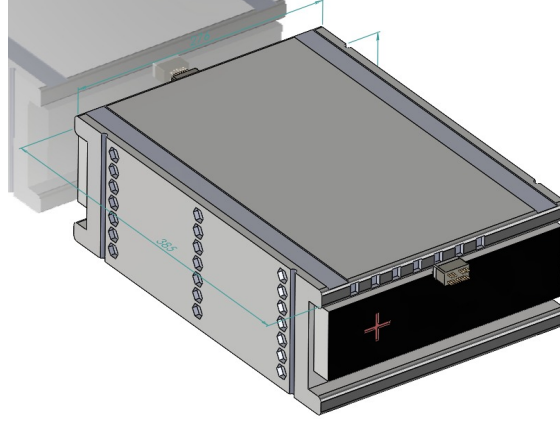


Figure 7: Battery module - Axonometric view

## 5 Gravimetric and volumetric energy density

The **gravimetric energy density** is a parameter that defines how much energy a system contains in relation to its mass. It has been calculated for both the cells and module levels for comparison. Similarly, the **volumetric energy density** defines the amount of energy contained in the system in relation to its volume. High values of these two specifications indicate that the battery pack is well-optimized. It is also important to compare the energy density of the  $11s2p$  cells configuration with that of the module to better understand whether it has been properly dimensioned.

Equation 3 and Equation 4 could be used to compute these two values. The results are related to the module energy density.

$$d_g = \frac{E}{Mass} = \frac{52.5 \text{ kWh}}{27.2 \text{ kg}} = 1.93 \text{ kWh/kg} \quad (3)$$

$$d_V = \frac{E}{Volume} = \frac{52.5 \text{ kWh}}{0.0141 \text{ m}^3} = 3723 \text{ kWh/m}^3 \quad (4)$$

In the following Table 5 the results for both cells and module are reported, in order to compare the dimensioning efficiency.

|       | Cells | Module |                    |
|-------|-------|--------|--------------------|
| $d_g$ | 2.48  | 1.93   | kWh/kg             |
| $d_V$ | 5108  | 3723   | kWh/m <sup>3</sup> |

Table 5: Energy density comparison between cells and module

Considering these results, it is possible to compute the efficiency of the module dimensioning by using the following formulas:

$$\eta_g = 1 - \frac{2.48 \text{ kWh/kg} - 1.93 \text{ kWh/kg}}{2.48 \text{ kWh/kg}} = 0.778 \quad (5)$$

$$\eta_V = 1 - \frac{5108 \text{ kWh/m}^3 - 3723 \text{ kWh/m}^3}{5108 \text{ kWh/m}^3} = 0.729 \quad (6)$$

## 6 Conclusion

In conclusion, the battery dimensioning for the Renault Zoe has been a challenging yet rewarding process. The design aimed to optimize the balance between performance, safety, and efficiency, focusing on factors like voltage, energy, and cell configuration. The intricacies of selecting the right cell type has been explored, opting for LG Chem’s E63 lithium-ion pouch cell, and the optimal layout, which is a 10-module series connection with a 11s2p configuration.

One of the main challenges was meeting the energy density goals while maintaining the structural integrity and thermal stability of the battery. Thermal management, in particular, was crucial due to the high operating temperatures of lithium-ion batteries. The use of a passive indirect liquid cooling system with a cold plate was selected to manage these temperatures, but further optimization could involve alternatives like graphite for better heat conduction.

Moreover, the volumetric and gravimetric energy densities, which are critical for vehicle range and efficiency, showed room for improvement. The system’s energy density could be enhanced by refining the battery layout, such as possibly eliminating the cooling plate to free up more space, though this would require advanced thermal management techniques.

The calculated efficiencies of the gravimetric and volumetric energy densities showed that there is potential for improvement, with values of 77.8% and 72.9%, respectively. Thus, while the current design meets the specifications, there are still avenues for optimization in terms of both mass and volume efficiency.

In summary, the design is a solid foundation for the battery pack of the Renault Zoe, but ongoing research and innovation in thermal management, material selection, and battery chemistry could make the system more efficient and cost-effective in future iterations.